

Feature Selection in Statistical Machine Learning

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Lecture 21

Recap and outline

- Shapley value framework
- Shapley values and feature selection
- Example

Shapley value axioms

- General framework for feature attribution measures that satisfy 4 properties/axioms

1. Efficiency

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset)$$

2. Symmetry

If X_j and X_k are interchangeable then

$$\phi_j = \phi_k$$

3. Dummy

If X_k contains no information about Y

$$\phi_k = 0$$

4. Additivity

If $f = f_1 + f_2$, for all j

$$\phi_j(f) = \phi_j(f_1) + \phi_j(f_2)$$

Shapley value general formula

Feature set $(X_1, X_2, \dots, X_p)$

$S \subset \{1, 2, \dots, p\}$ denotes any subset of indices

Value function:

$$v : 2^p \rightarrow \mathbb{R}$$

is a function over a subset of the features (will see various examples later)

Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1, 2, \dots, p\} / \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} (v(S \cup \{j\}) - v(S))$$

Provably satisfies all 4 properties

$$\frac{v(S \cup \{j\}) - v(S)}{2!}$$

Example choices of $v(\cdot)$ functional

Conditional Expectation:

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

- Local attribution (depends on local point x)
- How much does feature contribute to function $f(\cdot)$

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset) = f(x_1, x_2, \dots, x_p) - \mathbb{E}[f(X_1, X_2, \dots, X_p)]$$

Global risk (naturally connects to feature selection):

$$v(S) = -\mathbb{E}[\ell(f_S(X_S), Y)]$$

- Local attribution (depends on local point x)
- How much does feature contribute to risk difference

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset) = \mathbb{E}[\ell(f_\emptyset(X_\emptyset), Y)] - \mathbb{E}[\ell(f_{1,2,\dots,p}(X_{1,2,\dots,p}), Y)]$$

Connecting Shapley values and feature selection

- Connecting feature selection and Shapley values
- Use value function:

$$v(S) = -\mathbb{E}[\ell(f_S(X_S), Y)]$$

- Recall LOCO feature importance:

$$I_j = \mathbb{E}[\ell(Y, f_{-j}(X_{-j})) - \ell(Y, f(X))]$$

- Recall feature selection (feature-wise) criteria:

$$Y \perp X_j | X_{-j} \Leftrightarrow \text{Do not select } X_j \Leftrightarrow I_j = 0$$

- **Question:** For Shapley values is it true that:

$$\phi_j = 0 \Leftrightarrow Y \perp X_j | X_{-j}?$$

Shapley values and LOCO formula comparison

- LOCO:

$$I_j = \mathbb{E}[\ell(Y, f_{-j}(X_{-j})) - \ell(Y, f(X))]$$

- Shapley value:

$$\phi_j = \sum_{S \subset \{1, 2, \dots, p\} / \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} \mathbb{E}[\ell(Y, f_S(X_S)) - \ell(Y, f_{S \cup \{j\}}(X_{S \cup \{j\}}))]$$

- Now consider a more general subset-relative LOCO definition

$$I_j^S = \mathbb{E}[\ell(Y, f_S(X_S)) - \ell(Y, f_{S \cup \{j\}}(X_{S \cup \{j\}}))]$$

- I_j^S measures how much value feature X_j adds relative to the subset of features X_S

Shapley value and LOCO comparison cont.

Consider a more general subset-relative LOCO definition

$$I_j^S = \mathbb{E}[\ell(Y, f_S(X_S)) - \ell(Y, f_{S \cup \{j\}}(X_{S \cup \{j\}}))]$$

I_j^S measures how much value feature X_j adds relative to the subset of features X_S Shapley value:

$$\phi_j = \sum_{S \subset \{1, 2, \dots, p\} / \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} I_j^S$$

Shapley a weighted average of LOCO statistics across different subsets May be viewed as an average LOCO importance

Example: 2-feature structural equation model

$$Y = X_2 + \epsilon_Y$$

$$X_2 = X_1 + \epsilon_2$$

where $X_1 \sim \text{Normal}(0, \sigma_1^2)$, $\epsilon_2 \sim \text{Normal}(0, \sigma_2^2)$ and $\epsilon_Y \sim \text{Normal}(0, \sigma_Y^2)$ are independent random variables.

Goal: Find ϕ_1 and ϕ_2 where $v(S) = -\mathbb{E}[(Y - f_S(X_S))^2]$ and think about the question:

$$\phi_j = 0 \Leftrightarrow Y \perp X_j | X_{-j}?$$

Example: 2-feature structural equation model cont.

After calculations:

$$\phi_1 = \frac{1}{2}\sigma_1^2$$

$$\phi_2 = \frac{1}{2}\sigma_1^2 + \sigma_2^2$$

Question: For Shapley values is it true that:

$$\phi_j = 0 \Leftrightarrow Y \perp X_j | X_{-j}?$$

Answer on Shapley values

$$\phi_j = 0 \Rightarrow Y \perp X_j | X_{-j}$$

but

$$Y \perp X_j | X_{-j} \nRightarrow \phi_j = 0$$

Consider 2-feature example: $Y \perp X_1 | X_2$ but $\phi_1 \neq 0$

Why?

- Shapley values take an average feature importance across all subsets
- LOCO on the other hand only looks at importance given all other features
- From a DAG perspective, Shapley values capture both direct and indirect effects
- Feature selection by our conditional independence definition captures only direct effects
- Under some mild conditions

$$Y \perp X_j \Rightarrow \phi_j = 0$$

no condition

Relating back to condition 3 (Dummy)

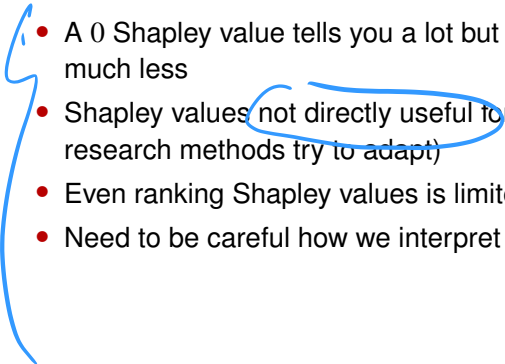
- **Dummy condition:** If X_k contains no information about Y

$$\phi_k = 0$$

- Since Shapley considers all possible subsets, Shapley will be non-zero if there is any direct or indirect effect on Y
- Under some mild conditions

$$Y \perp X_j \Rightarrow \phi_j = 0$$

Summary

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- A 0 Shapley value tells you a lot but a non-zero Shapley value says much less
 - Shapley values not directly useful for feature selection (recent research methods try to adapt)
 - Even ranking Shapley values is limited in value
 - Need to be careful how we interpret results